

# Carla - Maria Băraștean

Automation and Computer Science Graduate

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carlabarastean@gmail.com | +40 771 732 396

Cluj-Napoca, Romania | Portfolio

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## Hiring Team

Preh

Iași, Romania

## Application for System Control Engineer

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Dear Preh Hiring Team,

I am applying for the System Control Engineer position in Iași. I recently graduated in Automation and Computer Science from the Technical University of Cluj-Napoca with a final average of 9.25/10. This role is close to the reason I chose the field: I want to understand physical systems, model their dynamics, and design control logic that can be tested on real hardware.

During a six-month placement at Bosch, I worked on a six-component electric-vehicle drivetrain model using Simulink and ASCET, supported validation in TPT, generated C code, and deployed software on an STM32 microcontroller. The experience confirmed that this is the work I want to continue. I especially enjoyed understanding how each component influenced the overall dynamics and tracing results that did not match expectations until the model made physical sense.

My bachelor's thesis focused on nonlinear control and feedback linearization. I studied the theory and literature, then translated it into a practical MATLAB implementation for control analysis and design. Moving between differential equations, system dynamics, controller design, simulations, and software strengthened my mathematical foundation. It also made me certain that control engineering should remain at the center of my work, not only as a university subject.

I also have practical exposure to STM32 and Arduino-based systems, including sensor-driven and servo-actuated prototypes, so connecting control logic to electronics is familiar rather than abstract. Preh's description of an engineer who asks whether a solution is truly the best one felt familiar. I naturally compare alternatives and question assumptions, while remembering that an engineering result must also be delivered, documented, and understood by the team.

Regarding the on-site location, I currently live in Cluj-Napoca, but relocation is not an obstacle. I like Iași and would be happy to move there for a role that fits my direction this well. As a recent graduate, I am flexible, and I have considered master's programmes in both Cluj-Napoca and Iași, so my studies do not tie me to Cluj. I am prepared to relocate for the opportunity to work on control systems every day and learn directly from experienced engineers.

Thank you for considering my application. I would welcome the opportunity to discuss how my background in control theory, model-based development, and embedded implementation could contribute to Preh's engineering team.

Kind regards,

**Carla - Maria Băraștean**